LAB 07 ANSWERS REPORT

HANS PUJALTE (PUJA0009)

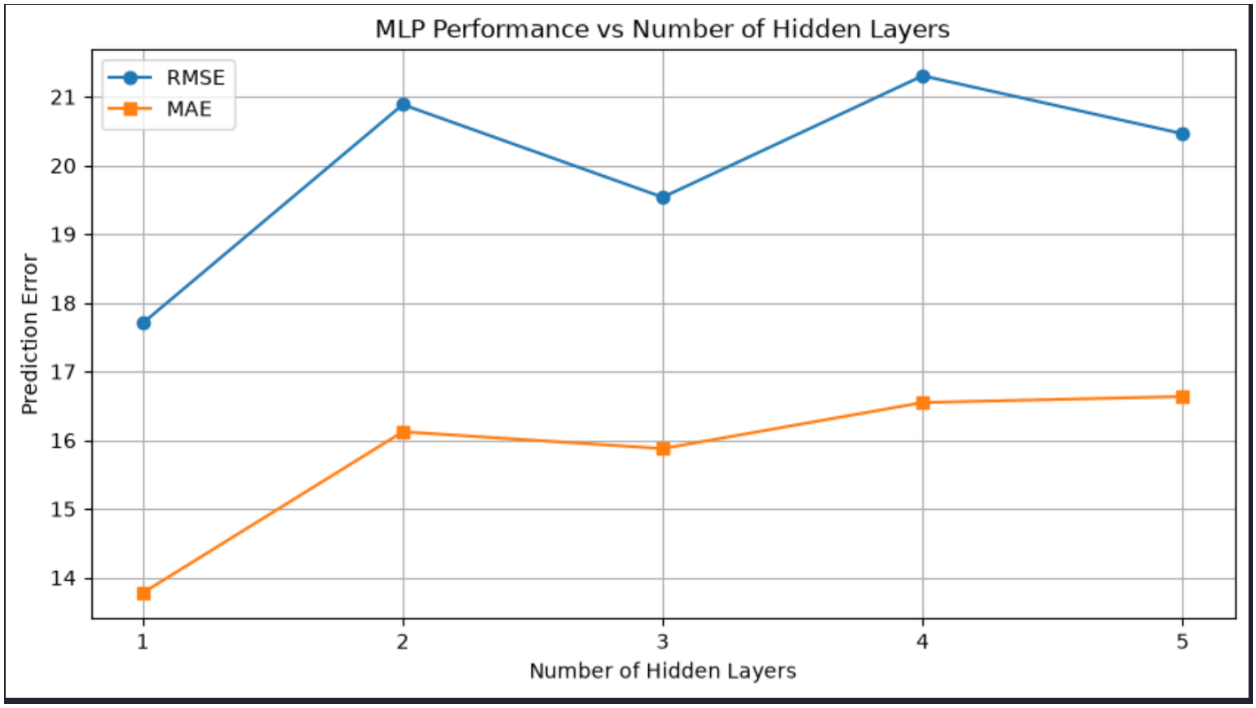
COMP3742, TONSLEY
COMPLETED: 19/10/2026

Hans Pujalte

Note: I submitted this lab extremely late but wanted to complete it anyway

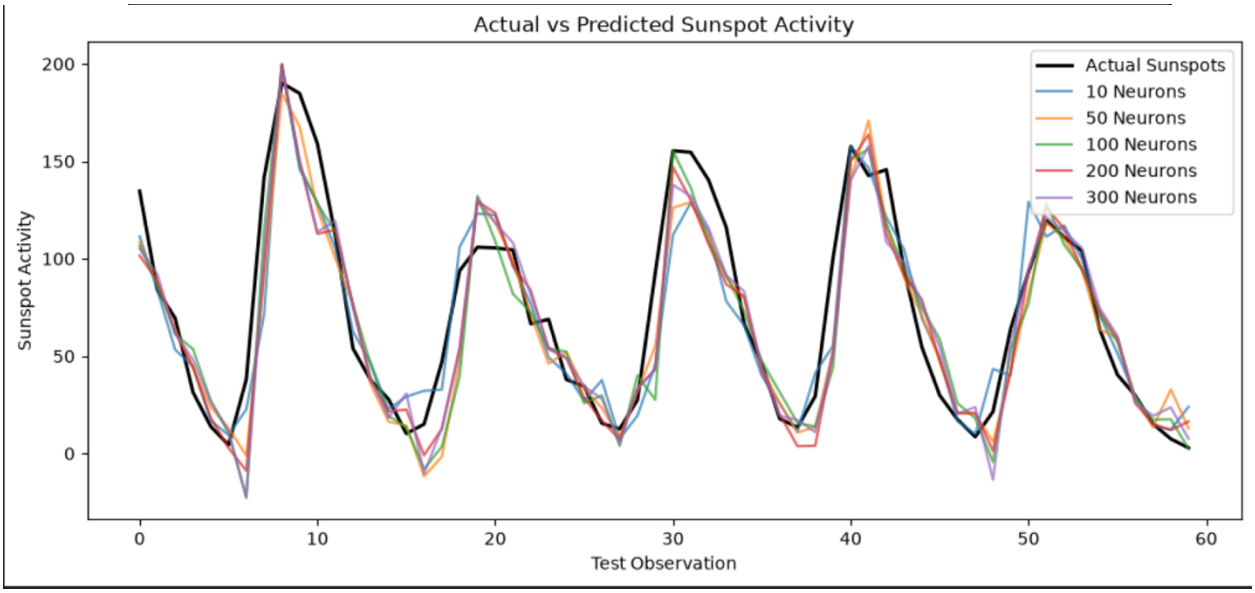
1. Using scikit-learn fit a MLP to the sunspot data using ReLU activation function and the adam solver. Evaluate the change in Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as the number of hidden layers is increased from 1 to 5 with 50 nodes per hidden layer. Plot and discuss the results.

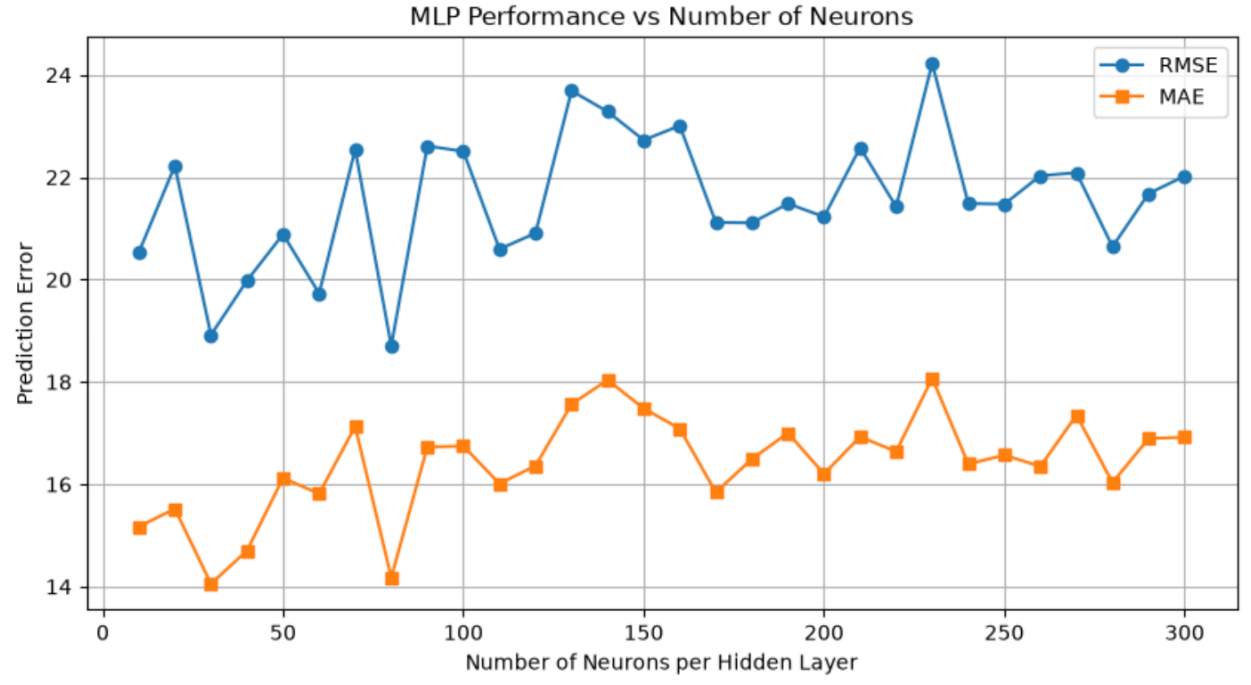
	Hidden Layers	RMSE	MAE
0	1	17.704217	13.776954
1	2	20.886130	16.122600
2	3	19.536947	15.877979
3	4	21.305167	16.548322
4	5	20.461445	16.635903



As the number of hidden layers increased, there was no significant change in RMSE or MAE. Prediction accuracy remained relatively the same, with minor fluctuations which suggest a simpler architecture may be sufficient to predict patterns in this dataset.

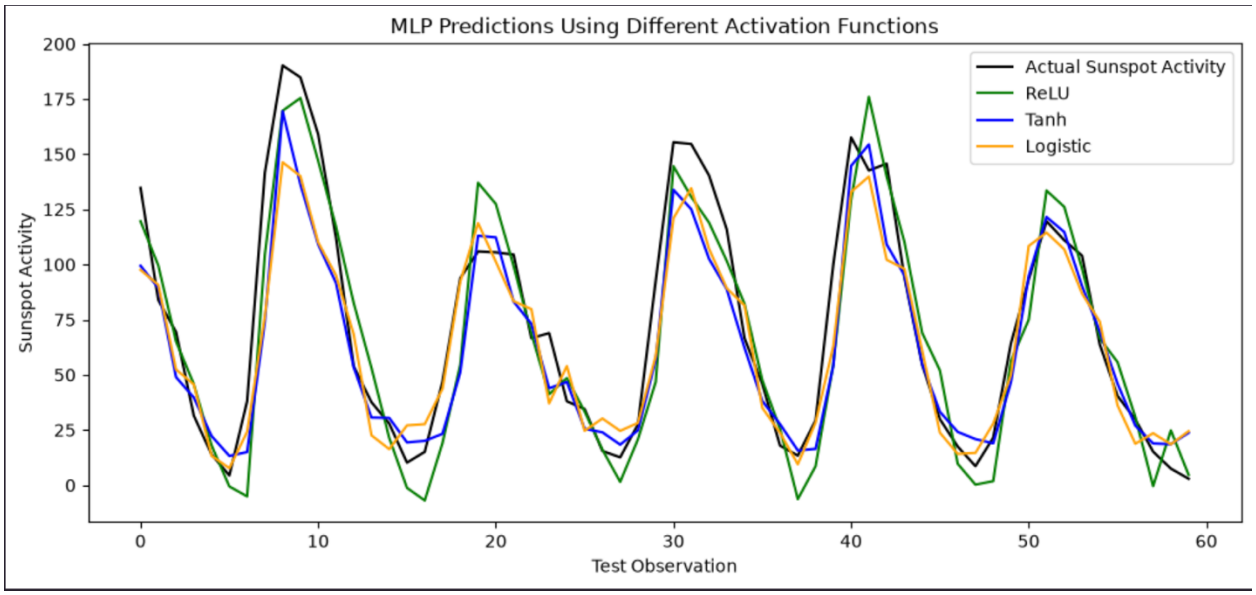
2. Using scikit-learn fit a 2 hidden layer MLP to the sunspot data using ReLU activation function and the adam solver. Evaluate the change in RMSE and MAE as the number of nodes in each layer is increased from 10 to 300. Plot and discuss the results.



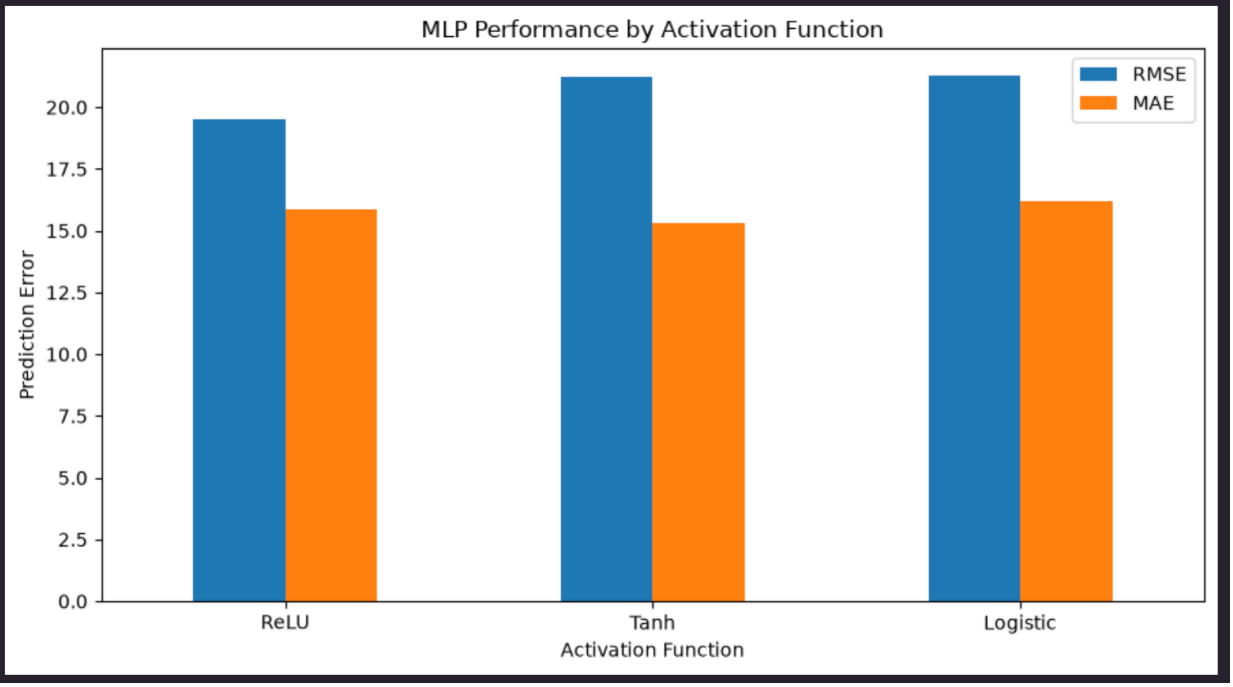


Similar to Question 1, the same findings were observed when the number of neurons increased in each hidden layer. Slight fluctuations were present, and prediction accuracy was relatively consistent when trained with models of different neurons.

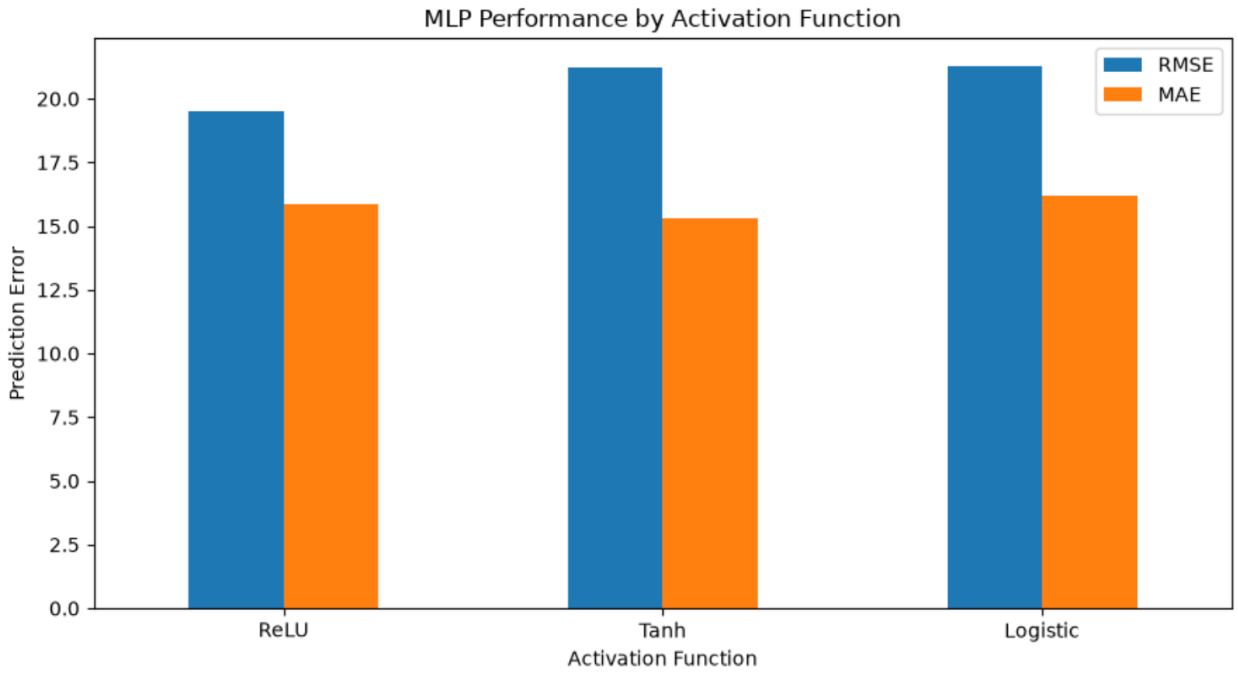
3. Using scikit-learn fit a 3 hidden layer MLP (50 hidden nodes per layer) to the sunspot data using ReLU activation function and the adam solver. Evaluate the change in RMSE and MAE using at least 3 different activation functions. Plot and discuss the results.



	Activation	RMSE	MAE
0	ReLU	19.536947	15.877979
1	Tanh	21.211985	15.322374
2	Logistic	21.304277	16.199416



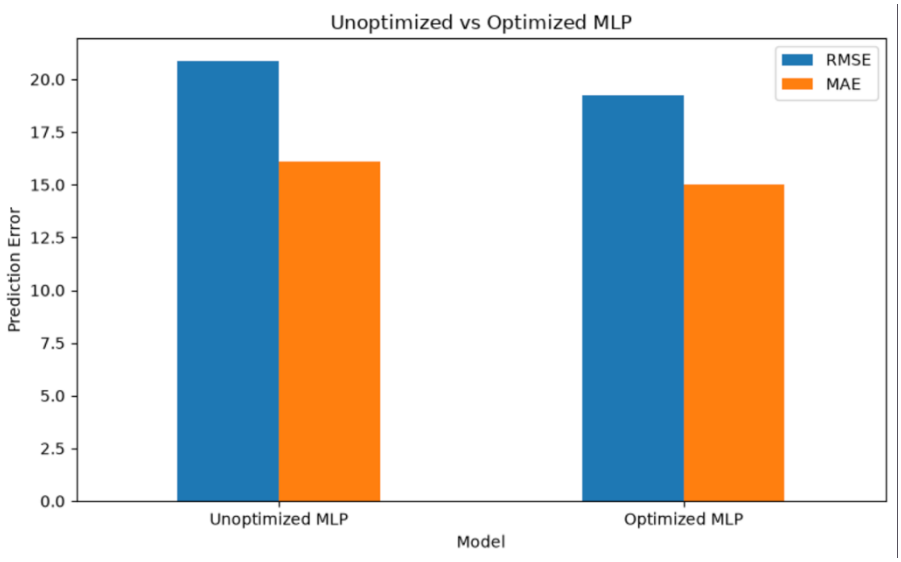
All 3 activation functions produced a RMSE and MAE that was consistent across all models. This suggest that the selected activation function for the sunspot dataset has limited influence over predictive performance.



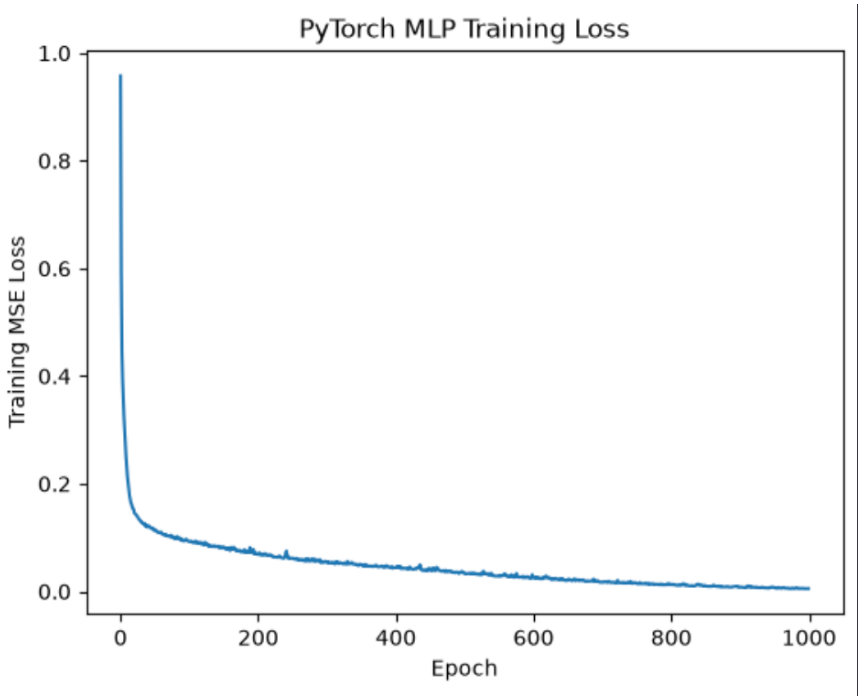
4. Using scikit-learn fit a 2 hidden layer MLP (50 hidden nodes per layer) to the sunspot data using ReLU activation function and the adam solver. Perform Grid Search hyperparameter optimization for different solvers, batch sizes, and learning rates using cross-validation. Compare RMSE, and MAE between the unoptimized and optimized models.

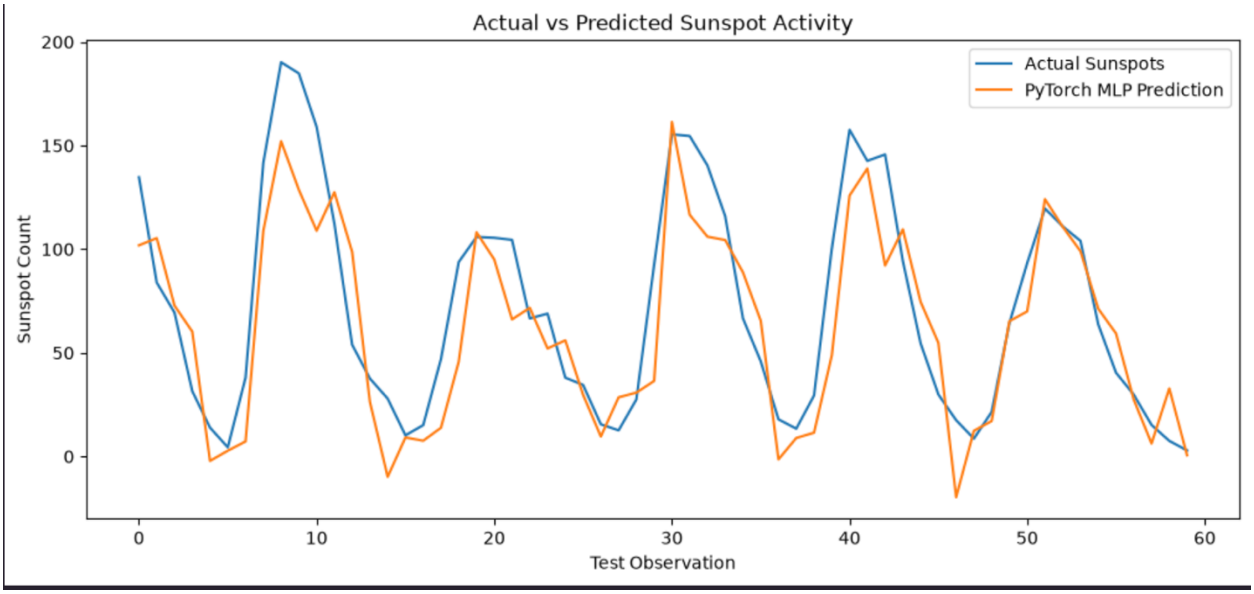
The unoptimised and optimised model produced a RMSE and MAE score that was consistent with each other. However, the optimized model showcased a slight improvement over its counterpart, with an RMSE of 19.23 and MAE of 15. This suggests that hyperparameter tuning was beneficial but did not provide a significant performance advantage.

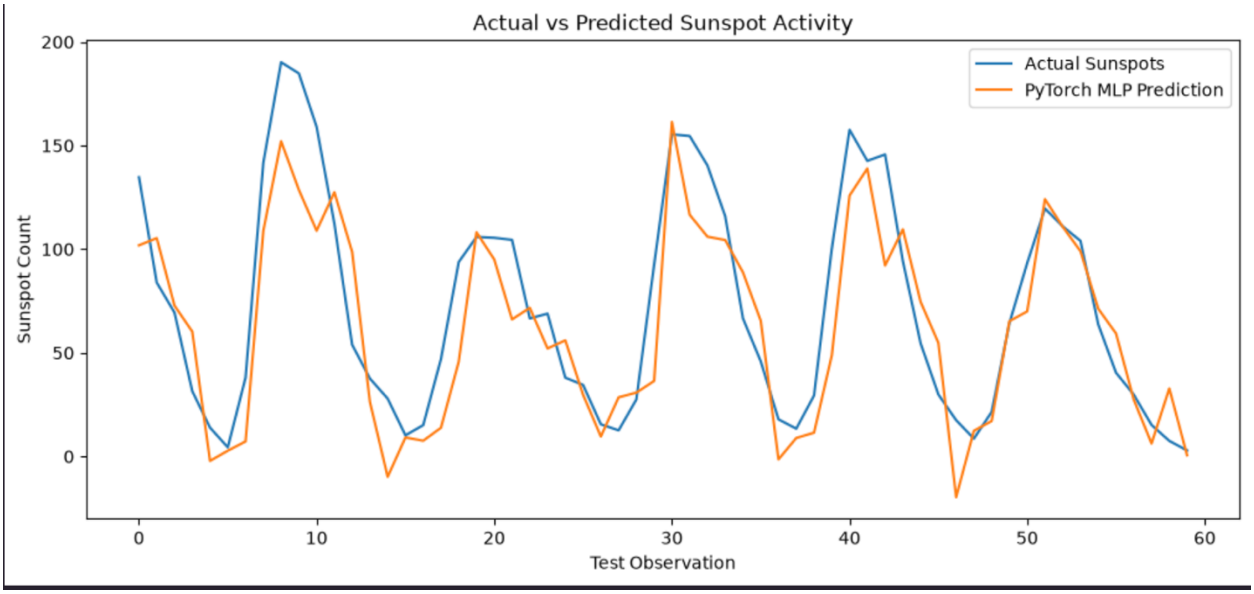
	Model	RMSE	MAE
0	Unoptimized MLP	20.886130	16.122600
1	Optimized MLP	19.230081	15.004235



5. Implement an MLP architecture (two hidden layers with 50 and 25 nodes respectively, both layers containing 'tanh' activation) using PyTorch. Train it on the same sunspot dataset and evaluate its performance.







As illustrated by the line plot, the model's MSE loss significantly decreased as the number of epochs was increased from initially 200 to 1000. The model achieved a RMSE score of 21.67 and a MAE of 15.79 which is a similar to the scores seen in previous questions.